

Senior Data Engineer with 10+ years of experience designing and scaling cloud-native data platforms, real-time streaming systems, and enterprise lakehouse architectures. Proven expertise in Apache Spark, Kafka, Databricks, and multi-cloud ecosystems (Azure, AWS, GCP) to deliver high-throughput, low-latency analytics solutions. Specialized in building fault-tolerant batch and streaming pipelines, optimizing big data performance, and enabling self-service BI through curated analytical models. Strong track record of modernizing legacy systems, improving data reliability and governance, and driving measurable business impact through cost optimization, operational efficiency, and real-time insights.

Technical skills

- **Programming & Query Languages**
Python, SQL (PostgreSQL, SQL Server, Oracle, T-SQL, PL/SQL), PySpark, Spark SQL, Scala (basic), Java (data processing), Shell Scripting
- **Big Data & Distributed Processing**
Apache Spark, Spark Structured Streaming, Spark Streaming, Hadoop, Hive, MapReduce (legacy migration), Azure HDInsight (Spark, Hive), Distributed Batch & Micro-Batch Processing
- **Streaming & Event-Driven Systems**
Apache Kafka, Kafka Connect, Kafka Producers/Consumers, Spark Structured Streaming, Event-Driven Architectures, Real-Time Data Pipelines, Exactly-Once Processing Semantics, Checkpointing & Backpressure Handling
- **Data Modeling & Warehousing**
Dimensional Modeling (Star & Snowflake), Slowly Changing Dimensions (SCD Type 1 & 2), Fact & Dimension Design, Data Vault Concepts, Lakehouse Architecture, Medallion Architecture (Bronze/Silver/Gold), Schema Design, Enterprise Data Warehousing, Analytical & Reporting Models
- **Cloud Data Platforms**
Azure (Databricks, ADLS Gen2, Synapse Analytics, Data Factory, Event Hubs, Azure Monitor, RBAC),
AWS (S3, EMR, Glue, Lambda, Redshift, IAM, CloudWatch),
GCP (BigQuery, Dataproc, GCS),
Databricks Lakehouse Platform, Snowflake
- **Data Integration & Orchestration**
Apache Airflow, Azure Data Factory, Kafka Connect, CI/CD for Data Pipelines (Jenkins), Workflow Automation, Hybrid Batch + Streaming Orchestration
- **Data Storage & Lakehouse Technologies**
Delta Lake, Azure Data Lake Storage (ADLS), Hive Tables, Parquet, ORC, Columnar Storage Formats, Partitioning & Optimization Strategies
- **Performance Optimization & Scalability**
Spark Performance Tuning, Cluster Autoscaling, Partition Pruning, Broadcast Joins, Adaptive Query Execution (AQE), Memory Optimization, Cost Optimization Strategies
- **Data Quality, Governance & Security**
Data Validation Frameworks, Streaming Data Quality Checks, Metadata Management, Audit Controls, Encryption at Rest & In Transit, Role-Based Access Control (RBAC), Compliance Reporting Support (Financial & Regulatory Pipelines)
- **BI & Analytics Enablement**
Power BI, Tableau, Azure Synapse Analytics, Self-Service Analytics Enablement, Semantic Modeling, Executive Dashboards, Near Real-Time Reporting
- **DevOps & Platform Engineering**
CI/CD Pipelines, Infrastructure Optimization, Monitoring & Observability (Kafka Lag Monitoring, Spark Metrics, Cloud Monitoring), Log Analytics, SLA Management

- **Databases & NoSQL Systems**
Oracle, SQL Server, PostgreSQL, MongoDB, Cassandra, Relational & Distributed Data Stores

Work History

Senior Data Engineer *Cygnnet.One, Princeton, NJ | Nov 2021 – Present*

- Modernized enterprise analytics by architecting a **lakehouse on Azure Databricks + Azure Synapse**, enabling unified batch and streaming workloads and reducing data latency from hours to minutes.
- Delivered high-throughput ingestion by building **Kafka-based streaming pipelines** processing millions of daily events, unlocking near real-time operational and compliance reporting.
- Accelerated real-time analytics by implementing **Spark Structured Streaming** for incremental processing of financial and transactional data, improving data freshness by 70%.
- Enabled scalable CDC ingestion by integrating **Kafka producers with Oracle and SQL Server sources**, eliminating nightly batch dependencies and improving downstream reliability.
- Improved pipeline stability by designing fault-tolerant **streaming ETL with exactly-once semantics**, reducing data reconciliation issues by 60%.
- Optimized big-data transformations using **Spark SQL and PySpark**, cutting end-to-end processing time by 30% on high-volume datasets.
- Strengthened regulatory reporting by implementing real-time data validation checks within **streaming pipelines**, ensuring accuracy for financial filings (e.g., FR-Y9C, 2052A).
- Reduced cloud costs by 20% by tuning **Databricks autoscaling and Spark cluster configurations** for mixed batch and streaming workloads.
- Improved observability by integrating **Kafka lag monitoring and Spark streaming metrics**, enabling proactive incident detection before SLA breaches.
- Streamlined orchestration by coordinating **Airflow and Azure Data Factory** to manage hybrid batch-and-stream workflows across environments.
- Accelerated analytics delivery by publishing curated, low-latency datasets to **Azure Synapse**, enabling self-service BI for business teams.
- Enhanced data security by enforcing **encryption and RBAC across streaming and lake storage**, aligning with enterprise security and audit standards.
- Increased cross-team velocity by partnering with analysts and data scientists to productionize streaming datasets that powered executive dashboards and alerts.

Senior Data Engineer *CDW, Plano, TX | Jan 2017 – Aug 2021*

- Transformed legacy big-data systems by migrating **MapReduce and Pig workloads to Apache Spark**, improving processing speed by 40% and lowering infrastructure costs.
- Introduced real-time data ingestion using **Kafka + Spark Streaming**, enabling faster visibility into customer behavior and operational metrics.
- Built scalable batch and micro-batch pipelines on **Azure HDInsight (Spark, Hive)**, supporting multi-terabyte daily data volumes.
- Reduced data availability delays by redesigning ETL workflows in **Azure Data Factory**, cutting batch windows by more than half.
- Enabled advanced analytics by preparing high-quality datasets through **Spark-based feature engineering**, supporting churn and forecasting models.
- Improved platform resilience by implementing **checkpointing and back-pressure handling in Spark Streaming**, minimizing data loss during peak loads.
- Delivered enterprise reporting by publishing curated datasets to **Power BI**, enabling leadership to track KPIs in near real time.
- Strengthened governance by standardizing data validation and audit controls across **Hive and Spark pipelines**, improving trust in analytical outputs.
- Expanded data platform capabilities by integrating **NoSQL stores (MongoDB, Cassandra)** for semi-structured and event-driven use cases.

- Increased deployment reliability by introducing **CI/CD with Jenkins** for Spark jobs and ETL workflows.
- Led and mentored a team of data engineers, accelerating delivery of big-data and streaming initiatives across multiple business units.

Data Engineer *Damco Solutions, South Burlington, VT | Sep 2013 – Dec 2016*

- Built large-scale data processing systems using **Apache Hadoop and Spark**, enabling analytics on billions of records across diverse domains.
- Designed early-stage streaming solutions using **Spark Streaming**, reducing reliance on slow, file-based batch ingestion.
- Improved data availability by developing automated ingestion pipelines with **Airflow and Spark SQL**, eliminating manual processing steps.
- Enhanced query performance by modeling analytical datasets in **Hive**, supporting both OLAP and reporting workloads.
- Increased data accuracy by implementing robust **data validation and cleansing logic in Python and SQL**, reducing downstream errors.
- Enabled cloud readiness by integrating on-premise systems with **Azure Data Lake Storage**, laying the foundation for future cloud migration.
- Supported advanced analytics by preparing machine-learning-ready datasets using **Spark MLlib workflows**.
- Improved operational efficiency by automating recurring workflows with **Oozie and Airflow**, reducing job failures and manual intervention.
- Delivered business insights through **Tableau and Power BI dashboards**, helping stakeholders identify trends and performance gaps.

Education

Bachelor of Science in Computer Science University of Dallas,
May, 2013